

International Islamic University Chittagong

Department of Computer Science and Engineering (CSE)

BSc in CSE, Final Lab Examination, Autumn 2025

Course Code: CSE-3528 Course Title: Compiler lab.

Full Marks: 50

Time: 2.30 Hour

Question Number 3 is mandatory for all Students to Solve. You have to Solve total three Problems.

No	Problems	Marking
1.	Write a program that takes a simple grammar rule of the form and computes FIRST(A) and FOLLOW(A). $A \rightarrow aB$ $B \rightarrow b$	10
2.	Write a program that reads a basic arithmetic expression with integers and checks if types are valid (only int allowed). If any non-integer token appears, print "Type Error" Test Case: 1 Input: $a = 5 + 10 * 2$ Output: Valid Test Case: 2 Input: $a = 5 + 11.5$ Output: Type Error	10
3.	Write a program that converts a simple infix expression to 3-address code. Test Case: Input: $a = b + c * d$ Output: $t1 = c * d$ $t2 = b + t1$ $a = t2$	10
4.	Write a program that converts an expression into quadruple form. Follow the given format: (op, arg1, arg2, result) Test Case 1: Input: $a = b + c$ Output: (+, b, c, a) Test Case 2: Input: $a = b + c * d$ Output: (*, c, d, t1) (+, b, t1, a)	10

$E \rightarrow E$

5.	Write a program that converts an expression into triples form. Test Case 1: Input: a = b + c * d Output: O: (*, c, d) I: (+, b, (O))	10
6.	Write a program to convert a basic infix expression to prefix polist. Test Case 1: Input: a + b * c Output: + a * b c	10
7.	Implement LL(1) using C++	10
8.	Project Presentation	10
9.	Viva	10

The End